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Improving Stock Market Predictions Using LSTM Based On MLP's Comparative Analysis

Yuvika Saini^{a)}, Aleem Ali^{b)}, and Ananshu Kukreja^{c)}

Department of Computer Science and Engineering, Chandigarh University, Punjab, India-140413

Corresponding Author: ^{a)} yuvikasaini16@gmail.com

^{b)} aleem.e12948@cumail.in

^{c)} ananshukukreja@gmail.com

Abstract. The stock market serves as a vital investment platform, attracting numerous financial investors due to its growing capitalization. However, predicting stock prices accurately has always been a challenging task, requiring advanced algorithmic techniques. This research paper presents an innovative approach to improve stock market predictions by leveraging the strengths of Long Short-Term Memory (LSTM) based on a comparative analysis with Multi-Level Perceptron (MLP) models. The study compares the predictive capabilities of Random Forest, MLP, and LSTM models, with MLP demonstrating superior accuracy compared to Random Forest in the initial analysis. Building upon this finding, LSTM is employed by incorporating MLP predictions to forecast future market trends. The proposed technique demonstrates reliable results by combining machine learning and deep learning approaches. The MLP analysis further enhances the suggested model, resulting in improved accuracy and reduced mean square error. The result highlights the potential of integrating LSTM and MLP models for enhanced stock market predictions, offering valuable insights for financial forecasting and opening avenues for further research and refinement in this field.

Keywords: Stock market, Long Short-Term Memory, Random Forest, Multi-Level Perceptron, Financial Forecasting

INTRODUCTION

Stock market research is always a popular topic for academics in the financial and technical disciplines because it is one of the primary industries to which investors devote their attention. The rise in popularity of stock trading over the past several years is largely due to technological advancements. Investors look for strategies and instruments that will boost profits while lowering risks [1]. However, due to its non-linear, dynamic, stochastic, and unreliable characteristics, Stock Market Prediction (SMP) is not an easy task. SMP is a type of time-series forecasting that rapidly analyses past data and predicts the values of future data. Analysts from a variety of fields have expressed concern about the ability to predict the financial markets. Given the volatility of the stock market and the link between investing psychology and market behavior, complexity and non-linearity are two of the key problems. Driving money off of stock trading is a crucial component of stock market forecasting [2]. The difficulty of forecasting the stock market is mostly due to the stock time series' behavior, which resembles randomness. Because of global digitization, SMP has entered a technological era. Machine learning in stock price prediction is used to discover patterns in data. The trend of stock values and the index can be predicted if the data obtained from stock values are effectively per-processed and appropriate algorithms are used. In stock market prediction systems, machine learning and deep learning approaches can help investors and traders through their decisions [3]. These techniques aim to detect and understand patterns among massive volumes of data automatically. The algorithms can deal with forecasting price changes to enhance trading tactics.

Machine learning models such as Random Forest Classifiers, Decision Tree were applied to stock market behavior before introducing the time series model in the frame. The movement of the stock price was analyzed as a function of time series and was resolved as a regression issue. However, because of its erratic behavior and extreme volatility, anticipating the precise stock price values was quite challenging by using machine learning techniques [4]. The method that is most frequently employed is artificial neural networks (ANNs) which are part of Deep Learning. Multi-Level Perceptron is one of the most frequently used ANN for time series data. To minimize the objective function of the network by adjusting the connection weights, the MLP uses the backpropagation algorithm, which is based on the steepest gradient descent method, during training. Artificial neural networks of the MLP variety are able to simulate intricate nonlinear interactions between input characteristics and output labels. It can learn and reflect complex patterns in the data because of its many layers of interconnected neurons. Random Forest, in contrast, is a collection of decision trees that uses straightforward threshold-based principles and may have trouble

detecting complex nonlinear patterns. The motive is to prove how easily predictions can be concluded using Deep Learning over Machine learning. One of the best RNN architectures is called long short-term memory (LSTM). In the network's hidden layer, the memory cell, a new computational unit introduced by LSTM takes the place of conventional artificial neurons. These memory cells enable networks to efficiently link memories and remote input across time, making them well-suited to understand the changing structure of data over time with a high capacity for prediction. The paper that we have presented includes predicting stock returns of the Apple dataset which is trained by using both the Random Forest Classifier and MLP. The comparative analysis demonstrates that MLP's forecasts are more accurate and that they can be utilized to further forecast market trends. We will forecast future market trends using LSTM which will use the predictions already made by MLP. The objective is to provide a brief comparison between a deep learning model and a machine learning model. We will further predict future market patterns using LSTM because it is known for its simple predictions of time series data. At the end of the introduction, the forthcoming section will provide a brief overview of the subsequent sections in the paper. Next section will delve into a comprehensive literature review, examining the existing research and scholarly works relevant to the topic at hand. Following the literature review, methodology will focus on employed in the study. It aims to provide a clear and detailed account of the approach and procedures used to gather and analyze the data. It outlines the research design, including the overall framework or theoretical perspective guiding the study. The focus shifts to presenting the results obtained from the study. This section aims to provide a clear and concise summary of the findings, based on the analysis of the collected data.

The last section serves as the conclusion of the paper, where the findings from the study are synthesized and interpreted. This section aims to provide a concise summary of the key insights and implications derived from the research.

LITERATURE REVIEW

These papers provide insights on how to use LSTM for forecasting future market trends and comparing the performance of Random Forest with MLP. They also offer information on data preprocessing, feature selection, hyperparameter tuning, and evaluation metrics used in the experiments. In recent years, researchers have shown interest in applying Long Short-Term Memory (LSTM) neural networks for forecasting future market trends. In [1], B. Nair et al., (2010), proposes a genetic algorithm optimized decision tree-SVM based stock market trend prediction system. The combination of decision trees and support vector machines (SVM) is optimized using a genetic algorithm to enhance the prediction accuracy of stock market trends. In [2], S. Lahmiri (2011) compares the performance of Probabilistic Neural Networks (PNN) and SVM for stock market trend prediction using economic and technical information. The study evaluates the effectiveness of these two algorithms in predicting stock market trends based on different types of input data. In [3] B. Nair et al., (2015) presented a decision tree-rough set hybrid system for predicting stock market trend in their paper. This study combines the decision tree algorithm with rough set theory to enhance the prediction accuracy of stock market trends. Whereas in [4], R. Choudhry and K. Garg (2018) compared various machine learning algorithms for stock price prediction using technical indicators. The study evaluates the performance of algorithms such as decision trees, random forests, SVM, and neural networks in predicting stock prices based on technical indicators. In [5], T. Huynh and D.S. Kim (2018) propose a two-step kernel-based approach for financial market forecasting. The study combines kernel principal component analysis (KPCA) and kernel support vector regression (SVR) to predict stock market returns. Also in [6], Hiransha M. et al. (2018) focused on NSE (National Stock Exchange) stock market prediction using deep learning models. The study explores the application of deep learning techniques, such as Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN), for predicting stock market trends.

In [7], Lim, S. C. et al. (2018) investigate the forecasting of stock market returns over multiple time horizons using financial news articles. The study explores the use of textual analysis and sentiment analysis of financial news articles to predict short-term and long-term stock market returns. In [8], Heaton, J. et al. (2019) discuss the application of deep learning in finance. The paper provides an overview of the various deep learning techniques, such as deep neural networks and recurrent neural networks, and their applications in financial prediction tasks, including stock market prediction. In [9], Zheng, Y. et al. (2019) proposed DeepAR, a probabilistic forecasting approach using autoregressive recurrent networks. The study focuses on time series forecasting, including stock market prediction, using deep learning models that capture the dependencies and uncertainties in the data. To note in [10], J. Wang et al. (2020) propose a financial time series forecasting method using an attention-based recurrent neural network. The model incorporates an attention mechanism to focus on relevant information within the time series data and improve the accuracy of financial forecasting tasks. In [11], S. Mehtab and J. Sen (2020) present a stock price prediction approach using deep learning models based on Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM). The study explores the combination of CNN and LSTM architectures for

improved stock price prediction accuracy. In [12], Jinghua Zhao et al. (2021) introduced a prediction model for stock price trends based on recurrent neural networks. The study focuses on utilizing the temporal dependencies within the stock price data to forecast future trends and provides a framework for accurate trend prediction. In [13], K. Liu. Et al. (2021) proposed an enhanced stock price prediction method by combining the Long Short-Term Memory (LSTM) model with online social networks. The study explores the incorporation of social media data to improve the prediction accuracy of stock prices. Also in [14], N. Parveen. Et al. (2020) discuss an IoT-based automatic vehicle accident alert system. The study focuses on developing a system that detects vehicle accidents using IoT technology and sends automatic alerts to relevant authorities, aiding in prompt response and assistance. In [15], Sachdeva S. and Aleem Ali (2022) explore the application of machine learning with digital forensics for attack classification in a cloud network environment. The study investigates the use of machine learning algorithms to classify different types of attacks in cloud networks, thereby improving security and detection capabilities. [16], J. Choi. Et al. (2023) propose a hybrid information mixing module for stock movement prediction. The study introduces a novel approach that combines different types of information, such as textual and numerical data, to enhance the accuracy of stock movement prediction. In [17], Li.X. et al. (2023) conduct a comparative analysis of machine learning algorithms for stock market prediction. The study compares the performance of various machine learning algorithms in predicting stock market trends, providing insights into their relative strengths and weaknesses.

METHODOLOGY

Machine learning and Deep Learning are understandably one of the most widely used systems which take a large amount of distinct raw data and can be combined to make appropriate inferences. We have used three machine learning algorithms in this paper to predict future stock market trends. Out of those three algorithms, Random Forest Classifier and Multi-Level Perceptron are trained to predict accurate historical data. From which MLP gives more apt results. The predictions made by MLP are fed to the LSTM algorithm and by using LSTM we will be predicting the future market trend of the next 30 days. Generating historical data is the most crucial step in order to predict a model. The dataset in the proposed model consists of stock prices from the past 5 years. It consists of 1258 entries and parameters listed in Table 1.

TABLE 1. Parameters of the Dataset

Parameters used	Denotation
Date	Time of stock price
Open	Opening share price
Close	Closing share price
High	Highest share value of the day
Low	Lowest share value of the day
Volume	Number of shares traded

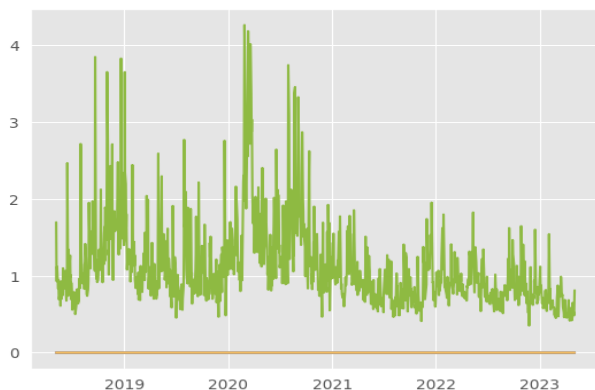


FIGURE 1. Graph of actual dataset

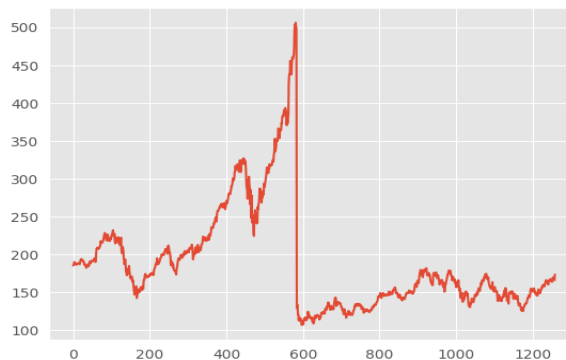


FIGURE 2. Opening and Closing graph of the dataset

The given flowchart mentioned in Fig. 3 shows the proposed methodology. According to the figure, we have identified the problem and the dataset corresponding to the problem. The data set of the last five years starting from 2018 is considered. In this paper, we would be applying Random Forest Classifier and MLP model to train the dataset and then use the LSTM model to predict the market trend of the next 30 days.

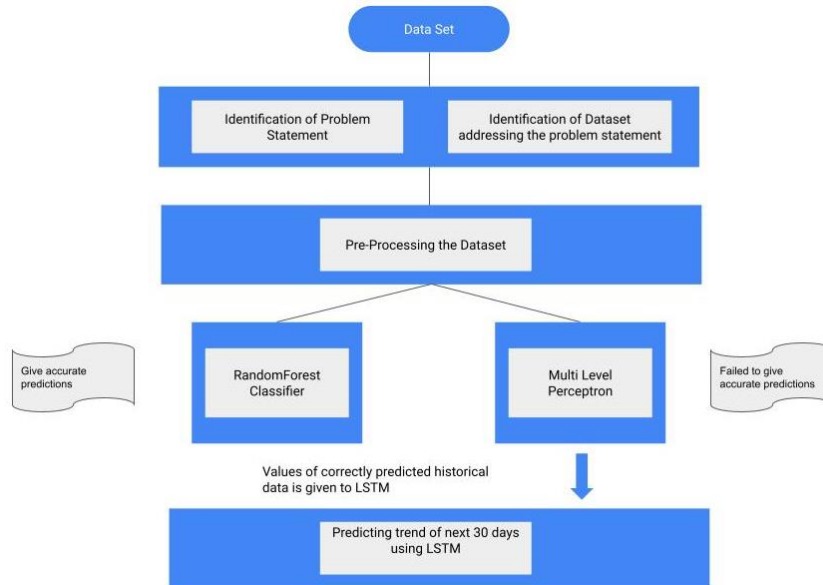


FIGURE 3. Flowchart of the methodology

The dataset collected is first trained and tested for both the models Random Forest Regressor and MLP separately to find that the predictions of MLP are more accurate. While Random Forest had an RMSE score of 0.52 which means it gave 52% accuracy, MLP gave a less RMSE score of 0.04 which showed less chance of error in the prediction. Since there was more use of hidden layers and hyperparameters in MLP more accurate result was determined. In Fig 4 we can see that the MLP model has quite efficiently predicted the actual dataset. Whereas in Fig 5, we can see the correct predictions made by Random Forest Classifier.

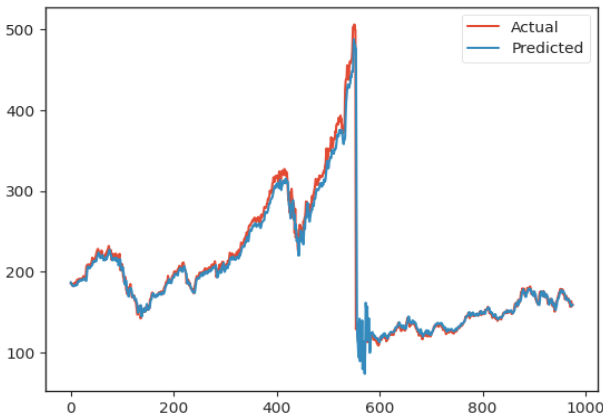


FIGURE 4. Predicted graph for MLP Model

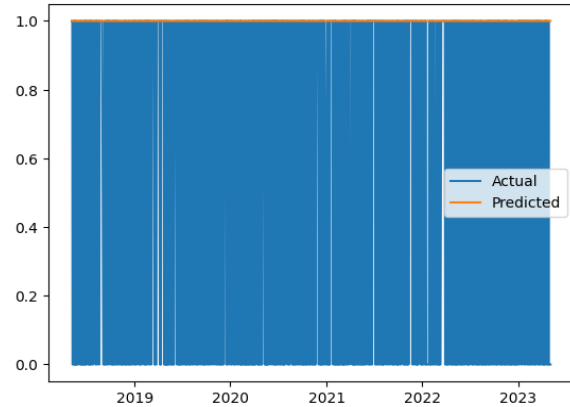


FIGURE 5. Predicted Graph of Random Forest Classifier Model

RESULT

The study and validation of the LSTM model for predicting future market trends using the predicted MLP results is the main topic of this section. By contrasting Fig. 6, we can observe that the LSTM model has correctly anticipated the trend for the upcoming 30 days by using the predicted training and testing dataset of MLP. Historical datasets from May 2018 to May 2023 were used for training and testing MLP algorithm as well as the Random Forest Classifier model which failed to give the required results. Training portions make up 80% of all datasets, and testing portions, 20%. Fig. 6 depicts the outcome predicted by the LSTM model. Fig. 7 gives a closer and more precise look at the predictions made for the next 30 days. LSTM networks are specifically created for sequential data processing, enabling them to better capture and predict the temporal dynamics evident in market movements. In Fig. 7, LSTM has provided a very smooth and precise result for every timestamp of the data. By efficiently capturing long-term dependencies, managing sequential data, overcoming gradient difficulties, and accepting variable-length sequences included in market data, LSTM networks along with MLP as a hybrid gave more accurate results than if an LSTM model with RandomForest Classifier were to be considered.

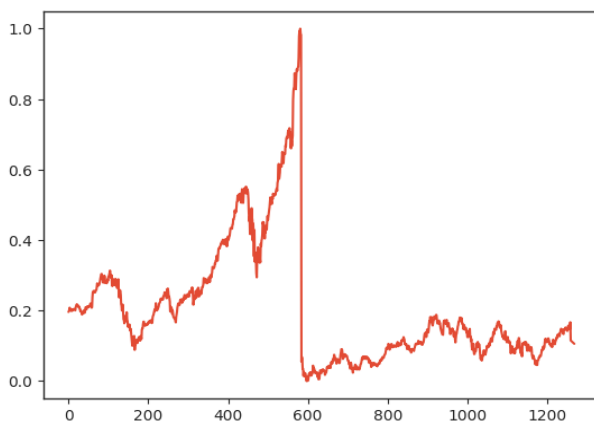


FIGURE 6. Predicted graph by LSTM model for next 30 days

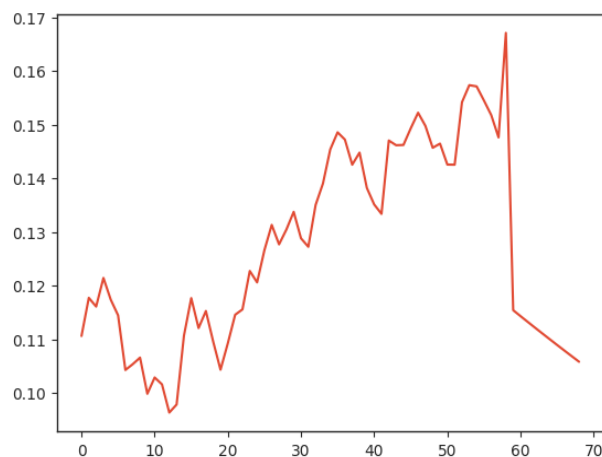


FIGURE 7. Depicts a closer look at Prediction by LSTM model for next 30 days

CONCLUSION

This study looked at how well the given historical data was predicted by Random Forest and Multi-Layer Perceptron (MLP) models. The study also used the projected values from the MLP model as input for an LSTM (Long Short-Term Memory) model to estimate future market patterns. The study's findings show that the MLP model gave a better and more advanced performance than the RandomForest Regressor. Due to its several hidden layers and non-linear activation functions, the MLP model also shows impressive prediction abilities. By feeding the projected values from the MLP model into the LSTM model, the research intended to capture the temporal dependencies and long-term patterns in the data in order to assess future market trends. The LSTM model, known for its ability to explain sequential data, showed promise in capturing the temporal dynamics of the dataset and provided insightful predictions for future market trends. In conclusion, this research demonstrates the potential effectiveness of combining Random Forest, MLP, and LSTM models for predicting market trends. The findings contribute to the field of financial forecasting and highlight avenues for further exploration and improvement in this domain.

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